A complete negative Einstein strictly almost-Kähler four-manifold: verification and a connection to Fino's existence question

Research continuation note

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The explicit metric and two-form below are the construction recorded by the user's research system, session A01, in Section 2 of `03_Subsequent_Mathematical_Developments.tex`. This note supplies an independent full curvature calculation, the simply connected complete extension, and an explicit connection to a published existence question. It does not attribute the construction to the present continuation.

1 The geometric statement

Let $G_0 = \mathrm{SL}(2,\mathbb{R})$ and let $K \cong \mathrm{SO}(2)$ be its maximal compact subgroup, with angular parameter of period 2π. Choose left-invariant forms such that

$$d\sigma_1 = -2\sigma_2 \wedge \sigma_3, \qquad d\sigma_2 = 2\sigma_3 \wedge \sigma_1, \qquad d\sigma_3 = 2\sigma_1 \wedge \sigma_2. \tag{1}$$

For $1/3 < x < 1$, put $D = 1 + 3x^2$ and

$$G = \frac{4D}{(1-x)^2(1+x)^2(3x-1)(3x+1)},$$

$$A_1 = \frac{4(3x-1)D}{3(1-x)(1+x)^2}, \qquad A_2 = \frac{4(3x+1)D}{3(1-x)^2(1+x)}, \quad A_3 = \frac{64x^2}{(1-x)(1+x)D}, \tag{2}$$

$$X_1 = \frac{(1-x)^3(3x+1)}{16x^3}, \qquad X_2 = \frac{(1+x)^3(3x-1)}{16x^3}. \tag{3}$$

Write

$$g = G\,dx^2 + A_1\sigma_1^2 + A_2\sigma_2^2 + A_3\sigma_3^2, \qquad e^0 = \sqrt{G}\,dx, \quad e^i = \sqrt{A_i}\sigma_i.$$

With $\omega_i = e^0 \wedge e^i + e^j \wedge e^k$ for cyclic (i,j,k), set

$$\Omega = \sqrt{X_1}\,\omega_1 + \sqrt{X_2}\,\omega_2. \tag{4}$$

Theorem 1. *The pair (g,Ω) extends smoothly over the zero section of*

$$E = G_0 \times_K \mathbb{R}^2 \longrightarrow G_0/K \cong \mathbb{H}^2,$$

where K acts with weight one on $\mathbb{R}^2$. The resulting Riemannian manifold is complete and diffeomorphic to $\mathbb{R}^4$. It satisfies

$$d\Omega = 0, \qquad |\Omega|_g^2 = 2, \qquad \mathrm{Ric}_g = -3g, \qquad W_g^- = 0,$$

and

$$|\nabla\Omega|^2 = \frac{36(1-x^2)^3}{(1+3x^2)^3} > 0 \tag{5}$$

at every point, including the zero section by continuous extension. In particular this is a strictly almost-Kähler Einstein structure. The metric itself is not homogeneous.

Here the orientation is the one induced by Ω, so that "self-dual" means $W^- = 0$. The scalar curvature is -12. Neither neutral signature nor a conformal change of an Einstein metric is involved.

2 Independent curvature and closure calculation

The calculation used here starts with the frame brackets and Koszul's formula; it does not assume the reduced Einstein or self-duality equations of the source manuscript. Put $dt = \sqrt{G}\,dx$, $v = dx/dt$, $h_i = vH_i$, and $w_i = vV_i$. Directly from (1)–(2),

$$\begin{aligned}
v^2 &= \frac{(1-x^2)^2(9x^2-1)}{4D},\\
\ell := \frac{v'}{v} &= -\frac{2x}{1-x^2} + \frac{9x}{9x^2-1} - \frac{3x}{D},\\
H_1 &= \frac{1}{2}\left(\frac{3}{3x-1} + \frac{6x}{D} + \frac{1}{1-x} - \frac{2}{1+x}\right),\\
H_2 &= \frac{1}{2}\left(\frac{3}{3x+1} + \frac{6x}{D} + \frac{2}{1-x} - \frac{1}{1+x}\right),\\
H_3 &= \frac{1}{x} + \frac{x}{1-x^2} - \frac{3x}{D},\\
V_1 &= -\frac{D}{4x(1+x)(3x+1)},\\
V_2 &= \frac{D}{4x(1-x)(3x-1)},\\
V_3 &= \frac{12x}{(9x^2-1)D}.
\end{aligned}$$

For the dual orthonormal frame, write $[E_a, E_b] = vc_{ab}^{\ d}E_d$. Its only independent nonzero coefficients are

$$c_{0i}^{\ i} = -H_i, \qquad c_{jk}^{\ i} = -2V_i \quad ((i,j,k) \text{ cyclic}).$$

If $\nabla_{E_a}E_b = v\gamma_{ab}^{\ d}E_d$, Koszul's formula gives

$$\gamma_{ab}^{\ d} = \tfrac{1}{2}(c_{ab}^{\ d} - c_{bd}^{\ a} + c_{da}^{\ b}). \tag{6}$$

With the convention $R(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z$, the full curvature follows from

$$\begin{aligned}
\langle R(E_a,E_b)E_c, E_d\rangle = v^2\Big[&\delta_{a0}(\gamma_{bc}^{\ d\prime} + \ell\gamma_{bc}^{\ d}) - \delta_{b0}(\gamma_{ac}^{\ d\prime} + \ell\gamma_{ac}^{\ d})\\
&+ \sum_e(\gamma_{bc}^{\ e}\gamma_{ae}^{\ d} - \gamma_{ac}^{\ e}\gamma_{be}^{\ d} - c_{ab}^{\ e}\gamma_{ec}^{\ d})\Big].
\end{aligned} \tag{7}$$

Rational substitution yields $\mathrm{Ric}_{ab} = -3\delta_{ab}$. In the orthonormal bases $(e^{0i} \pm e^{jk})/\sqrt{2}$, the curvature operator has zero mixed block and

$$\mathcal{R}^- = -\mathrm{Id}, \qquad \mathcal{R}^+ = \mathrm{diag}(r_1, r_2, r_3), \tag{8}$$

where

$$\begin{aligned}
r_1 &= -\frac{3(x-1)^2(21x^4+18x^3+12x^2-2x-1)}{D^3},\\
r_2 &= -\frac{3(x+1)^2(21x^4-18x^3+12x^2+2x-1)}{D^3},\\
r_3 &= \frac{9(x-1)^2(x+1)^2(5x^2-1)}{D^3}.
\end{aligned}$$

Their sum is -3. Thus $s = -12$ and $W^- = \mathcal{R}^- - (s/12)\mathrm{Id} = 0$.

For completeness, closure can be checked without curvature: if $f_i = \sqrt{X_i}$, the ith summand satisfies

$$d(f_i\omega_i) = [\dot{f}_i + (h_j + h_k - 2w_i)f_i]e^{0jk} = 0.$$

The derivative formula

$$v^{-1}(\nabla_{E_a}\Omega)_{bc} = \delta_{a0}\Omega'_{bc} - \sum_e (\gamma_{ab}^{\ e}\Omega_{ec} + \gamma_{ac}^{\ e}\Omega_{be})$$

then gives (5) by summing over a and $b < c$. The identity $X_1 + X_2 = 1$ gives $|\Omega|^2 = 2$. A self-dual two-form of length $\sqrt{2}$ defines an orthogonal almost-complex structure. Since it is closed and not parallel, that structure is strictly almost-Kähler. The accompanying SymPy script implements (6) and (7) directly and checks these identities with exact rational arithmetic.

3 Smooth extension and completeness

Define the geodesic radial coordinate r by $r = 0$ at $x = 1/3$ and $dr = \sqrt{G}\,dx$. The exact radial equation gives analytic functions of r^2 with expansions

$$\begin{aligned} x - \tfrac{1}{3} &= \tfrac{2}{9}r^2 - \tfrac{2}{81}r^4 + O(r^6), & A_1 &= r^2 + \tfrac{2}{9}r^4 + O(r^6), \\ A_2 &= 6 + 7r^2 + O(r^4), & A_3 &= 6 + 7r^2 + O(r^4), \\ A_2 - A_3 &= \tfrac{10}{3}r^4 + O(r^6), & X_2 &= \tfrac{8}{3}r^2 + O(r^4). \end{aligned}$$

Analyticity follows by first using $u = \sqrt{x - 1/3}$: r is an odd analytic function of u with nonzero linear coefficient, and hence has an odd analytic inverse.

Locally over G_0/K, take

$$\sigma_1 = d\psi + \beta, \qquad \sigma_2 + i\sigma_3 = e^{2i\psi}\zeta_0, \qquad z = re^{i\psi}.$$

The right K-action rotates the horizontal coframe with weight two. The normal circle has period 2π and weight one. Thus

$$dr + ir\sigma_1 = e^{-i\psi}(dz + iz\beta), \quad dr - ir\sigma_1 = e^{i\psi}(d\bar{z} - i\bar{z}\beta).$$

The leading normal metric is $|dz + iz\beta|^2$. Its correction has coefficient $O(r^4)\sigma_1^2$, which is smooth because $r^2\sigma_1 = \mathrm{Im}(\bar{z}\,dz) + r^2\beta$ is smooth. The horizontal trace is a smooth function of r^2 times $\sigma_2^2 + \sigma_3^2$. Its weight-four anisotropy has coefficient r^4 times a smooth even function, and is therefore smooth as well.

To check Ω, write $a_i = \sqrt{A_i}$, $a_1 = rb(r^2)$ and $x_i = \sqrt{X_i}$. Its mixed component is

$$\begin{aligned} &c_+ \,\mathrm{Re}((dr + ir\sigma_1) \wedge (\sigma_2 + i\sigma_3)) \\ &\quad + c_- \,\mathrm{Re}((dr - ir\sigma_1) \wedge (\sigma_2 + i\sigma_3)), \quad c_\pm = \tfrac{1}{2}x_2(a_2 \pm ba_3). \end{aligned}$$

Here $c_+ = rF(r^2)$ and $c_- = r^3H(r^2)$. The two expressions become

$$\mathrm{Re}[zF(r^2)(dz + iz\beta) \wedge \zeta_0], \qquad \mathrm{Re}[z^3H(r^2)(d\bar{z} - i\bar{z}\beta) \wedge \zeta_0],$$

which are smooth. The remaining radial–circle and horizontal area terms are smooth. These expressions describe the extension on the associated disk bundle, and are compatible under changes of local trivialization. The metric is positive definite at the bolt; self-duality and $|\Omega|^2 = 2$ continue there. At the bolt, $q = 32/3 > 0$.

For $\Gamma = \{1\}$ the base is the contractible hyperbolic plane. The oriented real plane bundle E is therefore trivial and $E \cong \mathbb{H}^2 \times \mathbb{R}^2 \cong \mathbb{R}^4$. To prove completeness, let $g_B = \sigma_2^2 + \sigma_3^2$ denote the

invariant complete metric on G_0/K. On every radial compact interval $0 \le r \le R$, the functions A_2, A_3 have a common positive lower bound c_R. Consequently

$$g \ge dr^2 + c_R \pi^* g_B \quad (0 \le r \le R).$$

In fact $A_2, A_3 \ge 6$ here, but the interval-dependent bound suffices. As $x \uparrow 1$, with $y = 1 - x$, one has

$$G \sim \frac{1}{2y^2}, \qquad A_1 \sim \frac{8}{3y}, \quad A_2 \sim \frac{32}{3y^2}, \quad A_3 \sim \frac{8}{y}.$$

Thus $r \to \infty$. A path of finite length has bounded r and bounded base distance by the displayed metric inequality. It remains in a disk bundle of bounded radius over a compact base ball, which is compact. No finite-length path escapes. Equivalently, closed metric balls are compact, and Hopf–Rinow proves completeness.

4 Nonhomogeneity and comparison with the classical ansatz

Since $s/12 = -1$, the eigenvalues of W^+ are $w_i = r_i + 1$. The intrinsic metric invariant, in curvature-operator norm, is

$$\mathrm{tr}((W^+)^2) = \frac{96(81x^{12} - 54x^{10} + 315x^8 - 168x^6 + 99x^4 - 18x^2 + 1)}{(1 + 3x^2)^6}. \tag{9}$$

At the bolt its value is $2/3$, whereas its limit at infinity is 6. It is nonconstant on the regular open set as well. The metric itself, and not merely the metric–form pair, is therefore nonhomogeneous. In particular it is not a left-invariant metric for any transitive Lie-group presentation.

The example also lies outside the weakly $*$-Einstein class treated by Armstrong's ansatz [2]. In the orthonormal self-dual basis $\omega_i/\sqrt{2}$, put $\sigma = \Omega/\sqrt{2}$. Directly,

$$\left|\mathrm{pr}_{\sigma^\perp}(W^+\sigma)\right|^2 = X_1 X_2 (w_1 - w_2)^2 = \frac{9(1 - x^2)^3(9x^2 - 1)}{(1 + 3x^2)^4} > 0. \tag{10}$$

Thus Ω is not a Weyl eigenform; the weakly $*$-Einstein condition in the Einstein almost-Kähler case fails. This is an explicit distinction from that classified subclass.

5 The precise published question answered

Fino [1], immediately after Corollary 2.1 in the published version, asks whether a real four-dimensional Lie group can admit a non-invariant, non-Ricci-flat Einstein strictly almost-Kähler metric. The question follows the nonexistence theorem for *left-invariant* Einstein strictly almost-Kähler structures. It contains no compactness or completeness requirement. The published paragraph is important: the 2003 arXiv version has the invariant nonexistence statement as Remark 2.2, but does not contain this subsequent question.

Corollary 2. *The metric–form pair in Theorem 1 supplies an affirmative answer to Fino's stated existence question. It supplies one even with the additional requirement of completeness, on the Lie group $(\mathbb{R}^4, +)$, with a non-left-invariant metric.*

Proof. Choose a diffeomorphism $\mathbb{R}^4 \to E$ and pull back the complete pair. The Einstein and almost-Kähler properties are invariant under diffeomorphism. Negative Ricci curvature excludes Ricci flatness, (5) excludes Kählerness, and the nonconstant invariant (9) excludes metric homogeneity. In particular the metric is not translation invariant on $(\mathbb{R}^4, +)$. □

There is an even shorter answer if completeness is not requested: the regular domain $(1/3, 1) \times \mathrm{SL}(2, \mathbb{R})$ is diffeomorphic to the four-dimensional Lie group $\mathbb{R} \times \mathrm{SL}(2, \mathbb{R})$. Pulling back the displayed formulas gives the required non-invariant Einstein strictly almost-Kähler metric there. This regular-domain metric is incomplete at the missing bolt; the complete example above uses a different underlying manifold.

Two distinctions fix the literature comparison. First, Apostolov–Drăghici's survey [3], Section 6, Remark 1, discusses the absence at that time of a negative-scalar self-dual strictly almost-Kähler Einstein four-dimensional example. The present example has exactly these properties. The *specific* question in that remark asks whether the canonical complex hyperbolic plane has such a local almost-Kähler structure. That specific question was answered negatively by Di Scala–Nagy [4], Corollary 1.1: it is not the question answered here, since (9) is nonconstant and this metric is not locally complex hyperbolic.

Second, the construction answers the stated existence question mathematically; the present comparison does not claim it is the first answer in the entire subsequent literature. An affirmative mathematical example and a claim of historical priority are different assertions.

The example is noncompact, with infinite volume. Thus it does not settle the closed negative-scalar Goldberg problem. The mathematical conclusion established here is a complete positive-definite four-dimensional example and its exact relationship to the recorded non-invariant existence question.

References

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